

Pipe Flow Pressure Loss Reference (U.S. Customary Units)

Section 1 — Major Losses (Darcy-Weisbach)

1.1 Darcy-Weisbach Equation

Head loss form:



$$h_f = f \cdot (L/D) \cdot (V^2 / 2g)$$

Symbol	Description	Units
h_f	Head loss (friction)	ft
f	Darcy friction factor (dimensionless) —	
L	Pipe length	ft
D	Pipe inside diameter	ft
V	Average flow velocity	ft/s
g	Gravitational acceleration	32.2 ft/s ²

Pressure drop form:



$$\Delta P = f \cdot (L/D) \cdot (\rho V^2 / 2) \cdot (1/144) \quad [\Delta P \text{ in psi}]$$

or equivalently, using specific weight γ (lbf/ft³):



$$\Delta P \text{ (psi)} = \gamma \cdot h_f / 144$$

where ρ is fluid density (lbm/ft³) and the 32.2 lbm·ft/(lbf·s²) conversion factor (g_c) is folded into the ρV^2 term if working in lbf.

1.2 Continuity / Velocity



$$Q = A \cdot V \quad V = Q / A \quad A = \pi D^2 / 4$$

Common unit conversion for liquid flow:





$$V \text{ (ft/s)} = 0.4085 \cdot Q \text{ (gpm)} / D^2 \text{ (in)}$$

(derived from $Q \text{ (cfs)} = Q \text{ (gpm)} / 448.831$, $A \text{ in ft}^2$)

1.3 Reynolds Number



$$Re = V \cdot D \cdot \rho / \mu = V \cdot D / \nu$$

Regime	Criterion
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Laminar	$Re < 2,300$
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Transitional	$2,300 < Re < 4,000$
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Turbulent	$Re > 4,000$
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1.4 Friction Factor

Laminar flow (exact):



$$f = 64 / Re$$

Turbulent flow — Colebrook equation (implicit, iterative):



$$1/\sqrt{f} = -2 \cdot \log_{10} [(\epsilon/D)/3.7 + 2.51/(Re \cdot \sqrt{f})]$$

Swamee-Jain explicit approximation ($\pm 1\%$ of Colebrook, no iteration, valid $5,000 < Re < 10^8$ and $10^{-6} < \epsilon/D < 10^{-2}$):



$$f = 0.25 / \{ \log_{10} [(\epsilon/D)/3.7 + 5.74/Re^{0.9}] \}^2$$

1.5 Typical Absolute Roughness (ϵ)

Pipe Material	ϵ (ft)	ϵ (in)
Drawn tubing / glass	0.000005	0.00006
Commercial/welded steel	0.00015	0.0018
Wrought iron	0.00015	0.0018
Asphalted cast iron	0.0004	0.0048
Galvanized iron	0.0005	0.006
Cast iron	0.00085	0.0102
Concrete	0.001 – 0.01	0.012 – 0.12
Riveted steel	0.003 – 0.03	0.036 – 0.36
PVC/plastic	0.000005	0.00006

1.6 Velocity Head



$$h_v = V^2 / 2g$$

Used as the common multiplier for both major and minor losses.

Section 2 — Minor Losses (Fittings, Valves, Entrances/Exits)

2.1 General Form

K-value (resistance coefficient) method:



$$h_m = K \cdot (V^2 / 2g)$$

Equivalent length method (alternative):



$$h_m = f \cdot (L_e/D) \cdot (V^2 / 2g)$$

Total system loss:



$$h_{L,total} = h_f + \Sigma h_m = [f \cdot (L/D) + \Sigma K] \cdot (V^2 / 2g)$$

2.2 K-Values — Fittings

Fitting	K (typical)
90° elbow, standard (threaded)	1.5
90° elbow, standard (flanged)	0.3
90° elbow, long radius (flanged)	0.2
45° elbow, standard	0.4
45° elbow, long radius	0.2
180° return bend (flanged)	0.2
Tee, flow through run (line flow)	0.2
Tee, flow through branch	1.0
Coupling / union	0.08
Pipe entrance, sharp-edged	0.5
Pipe entrance, rounded	0.04 – 0.05
Pipe entrance, re-entrant (Borda)	0.8 – 1.0
Pipe exit (into reservoir)	1.0

2.3 K-Values — Reducers / Expanders

Fitting	K (typical)
Gradual contraction ($\leq 45^\circ$ included angle)	0.04 – 0.08
Sudden contraction (based on D_2/D_1 , small ratio)	0.3 – 0.5
Gradual expansion (7° – 15° cone)	0.2 – 0.3
Sudden expansion	$K = [1 - (D_1/D_2)^2]^2$ (based on V_1 , upstream velocity)

2.4 K-Values — Valves (full open, unless noted)

Valve Type	K (typical)
Gate valve, full open	0.15 – 0.2
Gate valve, $\frac{3}{4}$ open	0.9 – 1.15
Gate valve, $\frac{1}{2}$ open	4.5 – 5.6
Globe valve, full open	6.0 – 10.0
Angle valve, full open	2.0 – 3.0
Ball valve, full bore, full open	0.04 – 0.1
Butterfly valve, full open	0.3 – 0.5
Plug valve, straightway	0.3 – 0.5
Swing check valve, full open	2.0 – 2.5
Lift check valve, full open	10 – 12
Diaphragm valve, full open	2.0 – 2.3

Note: valve K-values vary meaningfully by manufacturer, size, and internal geometry. For detailed engineering work, use the manufacturer's Cv data or Crane Technical Paper 410 ($2 \div K$ method) rather than generic table values.

2.5 Notes on Application

All K-values above assume fully turbulent flow and are referenced to the **velocity in the smaller/downstream pipe** unless otherwise noted.

For a piping system with multiple fittings of different sizes, K-values must be adjusted (scaled by $(D_{\text{small}}/D_{\text{large}})^4$) before summing, since K is diameter-dependent when comparing across different pipe sizes.

Cv-based valve sizing ($Q = C_v \cdot \sqrt{(\Delta P / SG)}$ for liquids, gpm/psi) is generally preferred over generic K-tables for control valves and vendor-specific equipment.